Project Title: Malaria Detection using Deep Learning

Project Design Phase-I - Solution Fit

Define CS, Fit into CC

1. CUSTOMER SEGMENT(S) — CS

Who is your customer?

- Lab Technicians, Healthcare Professionals
- Diagnostic Centers, Hospitals

6. CUSTOMER CONSTRAINTS — CC

What constraints prevent customers from choices?

- Time-Consuming Manual Diagnosis
- Human Error, Heavy Workload
- Limited Resources, No Automation

5. AVAILABLE SOLUTIONS — AS

What is already available in market?

- Manual Microscopy
- Traditional Diagnosis
- Limited AI Tools

Need: Automated, Accurate, Fast Solution

Explore AS, differentiate

Focus on J&TBD, Mid BE, Emotional TR

2. JOBS-TO-BE-DONE / PROBLEMS — J&TBD

Which job needs to be done?

- Fast & Accurate Malaria Diagnosis
- Reduce Manual Effort & Errors
- Automated Report Generation

9. PROBLEM ROOT CAUSE — RC

Why problem exists?

- Manual Observation
- Less Accuracy
- Time Pressure, Workload

7. BEHAVIOUR — BE

How do customers act today?

- Microscope Analysis
- Delayed Reports
- High Stress, Errors

Goal: AI-Based Automated System

Identify, Strong TR & Emotions

3. TRIGGERS — TR

What triggers the need?

- High Patient Load
- Diagnostic Delays
- Need for Accuracy

4. EMOTIONS: BEFORE / AFTER — EM

How do they feel?

Before: Stressed, Errors, Delay → After: Fast, Accurate, Reliable

10. YOUR SOLUTION — SL

Proposed Solution:

AI-Based Malaria Detection System

- Image Upload → AI Processing
- Prediction (Parasitized/Uninfected)
- Confidence Score + Instant Report

8. CHANNELS OF BEHAVIOUR — CH

- **B1. Online:** Web App (Flask)
- **B2. Offline:** Lab Integration
- **B3. Future:** Mobile App, Cloud DB

Identify, Strong TR & Emotions